

Supporting Information

Supporting Information for “Foundation or Formula? A Simulation-Based Comparison of Google’s TimesFM-3 and Classical Time-Series Models”

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Summary

This file contains supporting information referred to in the main text as Sections S1 to S5.

S1 | Results under the mean absolute percentage error

Table S1 reports mean MAPE by process, alongside the percentage of series on which it cannot be computed. The D8 row illustrates the problem: every intermittent series contains a zero, so MAPE is undefined for all of them, and the process on which TimesFM-3 achieves its largest and most consistent advantage simply drops out of the comparison. The rankings also move. On D4, MAPE places seasonal naive first (2.360) and TimesFM-3 second (2.801), whereas MASE places AutoARIMA first at every length from $n = 48$. MAPE's asymmetry between over- and under-forecasts is enough to reorder the methods on the same forecasts.

TABLE S1 | Mean MAPE over $h = 1, \dots, 12$ and percentage of series on which MAPE is undefined.

DGP	% undefined	Seas. naive	Theta	AutoETS	AutoARIMA	Combin.	TimesFM-3
D1	0%	2.785	2.737	2.681	2.336	2.501	2.283
D2	0%	3.291	3.341	3.277	2.696	2.971	2.641
D3	0%	9.509	9.278	9.875	10.078	9.393	9.214
D4	0%	2.360	7.471	4.986	2.862	4.538	2.801
D5	0%	2.971	3.583	3.270	2.436	2.916	2.256
D6	0%	2.011	2.379	1.835	1.902	1.925	1.822
D7	0%	7.634	4.631	2.979	2.538	2.767	2.184
D8	100%	—	—	—	—	—	—
D9	0%	1.861	1.769	1.741	1.470	1.599	1.457

Note: On the intermittent process D8 every series contains a zero, so MAPE cannot be computed for it. The best value in each row is in bold.

S2 | Monte Carlo standard errors

Table S2 gives the mean MASE of the six methods of the main design in every cell, with its Monte Carlo standard error.

S3 | Coverage by process

Table S3 gives the empirical coverage of the nominal 80% intervals of the four foundation-model configurations and of AutoARIMA and AutoETS, by process, and Table S4 compares the Gaussian and split-conformal intervals of the two classical models at $n \geq 96$.

S4 | Response-surface regressions

Tables S5 and S6 report the regressions summarised in the section of the main text on randomised parameters and departures from the assumptions.

TABLE S2 | Mean MASE and Monte Carlo standard errors, main design.

DGP	n	Seasonal naive	Theta	AutoETS	AutoARIMA	Combination	TimesFM-3
D1	24	1.646 (0.056)	1.682 (0.055)	1.568 (0.050)	1.485 (0.048)	1.538 (0.048)	1.417 (0.042)
D1	48	1.625 (0.058)	1.586 (0.055)	1.570 (0.054)	1.415 (0.043)	1.478 (0.048)	1.381 (0.039)
D1	96	1.626 (0.054)	1.572 (0.053)	1.564 (0.053)	1.324 (0.039)	1.435 (0.044)	1.307 (0.033)
D1	200	1.615 (0.061)	1.576 (0.057)	1.576 (0.057)	1.266 (0.035)	1.417 (0.046)	1.254 (0.033)
D2	24	1.905 (0.066)	2.003 (0.073)	1.886 (0.069)	1.708 (0.087)	1.788 (0.065)	1.675 (0.062)
D2	48	1.838 (0.068)	1.860 (0.069)	1.834 (0.068)	1.575 (0.053)	1.688 (0.058)	1.518 (0.042)
D2	96	1.810 (0.057)	1.812 (0.057)	1.811 (0.057)	1.517 (0.038)	1.643 (0.045)	1.455 (0.035)
D2	200	1.922 (0.067)	1.919 (0.067)	1.921 (0.067)	1.374 (0.034)	1.654 (0.050)	1.383 (0.033)
D3	24	4.411 (0.213)	4.483 (0.219)	4.892 (0.235)	5.275 (0.256)	4.659 (0.224)	4.656 (0.236)
D3	48	4.252 (0.208)	4.179 (0.206)	4.547 (0.207)	4.679 (0.239)	4.313 (0.203)	4.274 (0.205)
D3	96	4.125 (0.191)	4.135 (0.189)	3.938 (0.179)	3.940 (0.179)	3.871 (0.172)	4.023 (0.177)
D3	200	4.512 (0.210)	4.427 (0.211)	4.275 (0.199)	4.195 (0.201)	4.199 (0.199)	4.297 (0.198)
D4	24	1.029 (0.026)	1.146 (0.110)	3.602 (0.101)	2.284 (0.079)	3.084 (0.072)	1.321 (0.031)
D4	48	1.021 (0.024)	1.944 (0.048)	1.720 (0.032)	1.014 (0.030)	1.395 (0.028)	1.295 (0.027)
D4	96	1.005 (0.024)	2.910 (0.078)	2.407 (0.055)	0.886 (0.020)	1.839 (0.040)	1.124 (0.022)
D4	200	1.053 (0.022)	4.155 (0.117)	1.085 (0.024)	0.886 (0.021)	1.700 (0.043)	1.161 (0.020)
D5	24	1.139 (0.038)	3.154 (0.102)	3.025 (0.093)	1.574 (0.043)	2.474 (0.069)	1.152 (0.038)
D5	48	1.108 (0.032)	0.956 (0.038)	0.741 (0.027)	0.717 (0.024)	0.731 (0.023)	0.815 (0.029)
D5	96	1.076 (0.033)	0.757 (0.025)	0.761 (0.029)	0.732 (0.027)	0.713 (0.027)	0.761 (0.029)
D5	200	1.040 (0.030)	0.759 (0.020)	0.681 (0.024)	0.626 (0.022)	0.645 (0.022)	0.689 (0.024)
D6	24	0.801 (0.025)	1.327 (0.038)	0.766 (0.024)	0.802 (0.025)	0.837 (0.025)	0.807 (0.036)
D6	48	0.947 (0.028)	1.115 (0.031)	0.869 (0.023)	0.925 (0.027)	0.924 (0.025)	0.825 (0.022)
D6	96	1.006 (0.028)	0.963 (0.026)	0.905 (0.023)	0.920 (0.023)	0.916 (0.023)	0.865 (0.021)
D6	200	0.934 (0.023)	0.840 (0.017)	0.825 (0.017)	0.833 (0.017)	0.827 (0.017)	0.844 (0.019)
D7	24	5.417 (0.080)	3.179 (0.058)	1.272 (0.042)	1.083 (0.037)	1.680 (0.048)	1.142 (0.033)
D7	48	4.243 (0.067)	2.262 (0.048)	1.602 (0.061)	1.076 (0.051)	0.970 (0.038)	0.751 (0.022)
D7	96	1.655 (0.042)	0.821 (0.020)	1.305 (0.047)	1.329 (0.048)	1.097 (0.036)	0.959 (0.032)
D7	200	0.917 (0.026)	1.301 (0.031)	0.966 (0.028)	0.994 (0.030)	0.954 (0.027)	0.933 (0.025)
D8	24	1.019 (0.054)	0.999 (0.034)	1.047 (0.058)	0.996 (0.032)	1.010 (0.039)	0.780 (0.031)
D8	48	1.066 (0.066)	0.984 (0.036)	0.988 (0.033)	0.962 (0.024)	0.974 (0.028)	0.708 (0.030)
D8	96	0.932 (0.047)	0.923 (0.017)	0.926 (0.015)	0.924 (0.015)	0.923 (0.015)	0.683 (0.022)
D8	200	1.005 (0.052)	0.943 (0.017)	0.939 (0.015)	0.939 (0.015)	0.940 (0.015)	0.694 (0.023)
D9	24	1.547 (0.061)	1.518 (0.057)	1.454 (0.057)	1.372 (0.057)	1.404 (0.054)	1.321 (0.049)
D9	48	1.588 (0.056)	1.486 (0.054)	1.476 (0.053)	1.244 (0.041)	1.353 (0.046)	1.239 (0.039)
D9	96	1.430 (0.051)	1.342 (0.048)	1.333 (0.048)	1.098 (0.034)	1.213 (0.040)	1.104 (0.034)
D9	200	1.577 (0.056)	1.503 (0.053)	1.504 (0.053)	1.190 (0.032)	1.341 (0.042)	1.200 (0.035)

Note: Mean MASE over $h = 1, \dots, 12$ with its Monte Carlo standard error in parentheses (standard deviation over the 200 replications divided by $\sqrt{200}$).

TABLE S3 | Coverage of 80% prediction intervals by process.

Process	TimesFM-3	TimesFM-3 (eval.)	TimesFM-2.5	Chronos-Bolt	AutoARIMA	AutoETS
D1	0.817	0.819	0.787	0.814	0.777	0.852
D2	0.817	0.818	0.782	0.824	0.766	0.874
D3	0.704	0.707	0.703	0.721	0.706	0.618
D4	0.846	0.856	0.810	0.769	0.756	0.661
D5	0.818	0.830	0.799	0.833	0.724	0.686
D6	0.859	0.873	0.913	0.921	0.986	0.987
D7	0.845	0.848	0.894	0.968	0.833	0.765
D8	0.818	0.869	0.835	0.846	0.836	0.843
D9	0.807	0.809	0.783	0.810	0.760	0.825

Note: Empirical coverage of the nominal 80% interval $[q_{0.1}, q_{0.9}]$, averaged over lengths and replications; series-cluster bootstrap Monte Carlo SE at most 0.016.

TABLE S4 | Gaussian versus split-conformal 80% intervals for AutoARIMA and AutoETS.

Interval	D1	D2	D3	D4	D5	D6	D7	D8	D9
AutoARIMA, Gaussian	0.788	0.780	0.771	0.791	0.751	0.977	0.844	0.840	0.783
AutoARIMA, conformal	0.613	0.591	0.614	0.605	0.601	0.654	0.651	0.614	0.602
AutoETS, Gaussian	0.870	0.890	0.658	0.671	0.693	0.979	0.826	0.842	0.863
AutoETS, conformal	0.597	0.589	0.616	0.525	0.595	0.605	0.662	0.616	0.586

Note: Empirical coverage of the nominal 80% interval at $n \in \{96, 200\}$. Conformal: split-conformal intervals with two 12-step calibration windows (`statsforecast Conformal Intervals`); Gaussian: the models' analytic intervals.

TABLE S5 | Response-surface regressions of the robustness study, processes D1–D4.

Process	Term	Coefficient	95% interval	<i>p</i>	<i>R</i> ²
D1	version: est period	0.000	[-0.027, 0.027]	0.995	0.003
D1	version: heavy tail	-0.022	[-0.053, 0.009]	0.160	0.003
D1	version: outliers	-0.010	[-0.039, 0.019]	0.488	0.003
D1	$\log_2 n$	0.004	[-0.005, 0.014]	0.361	0.003
D1	parameter phi	-0.013	[-0.024, -0.001]	0.028	0.003
D1	parameter sigma	0.006	[-0.005, 0.016]	0.273	0.003
D2	version: est period	-0.000	[-0.027, 0.027]	0.991	0.012
D2	version: heavy tail	-0.016	[-0.049, 0.016]	0.328	0.012
D2	version: outliers	0.004	[-0.024, 0.032]	0.784	0.012
D2	$\log_2 n$	-0.002	[-0.011, 0.008]	0.724	0.012
D2	parameter phi	-0.031	[-0.043, -0.019]	0.000	0.012
D2	parameter sigma	0.010	[-0.001, 0.021]	0.063	0.012
D2	parameter theta	0.005	[-0.006, 0.015]	0.364	0.012
D3	version: est period	0.000	[-0.067, 0.068]	0.992	0.006
D3	version: heavy tail	0.020	[-0.049, 0.088]	0.574	0.006
D3	version: outliers	-0.004	[-0.070, 0.063]	0.915	0.006
D3	$\log_2 n$	0.014	[-0.007, 0.036]	0.192	0.006
D3	parameter phi	-0.046	[-0.072, -0.020]	0.000	0.006
D3	parameter sigma	-0.004	[-0.028, 0.020]	0.751	0.006
D3	parameter theta	0.017	[-0.007, 0.041]	0.164	0.006
D3	parameter drift	-0.010	[-0.034, 0.013]	0.398	0.006
D4	version: est period	-0.004	[-0.058, 0.051]	0.894	0.255
D4	version: heavy tail	0.061	[0.002, 0.121]	0.044	0.255
D4	version: outliers	-0.219	[-0.274, -0.164]	0.000	0.255
D4	$\log_2 n$	0.258	[0.240, 0.276]	0.000	0.255
D4	parameter phi	-0.055	[-0.075, -0.034]	0.000	0.255
D4	parameter sigma	0.047	[0.025, 0.068]	0.000	0.255
D4	parameter Phi	0.110	[0.089, 0.131]	0.000	0.255
D4	parameter amp	-0.038	[-0.058, -0.019]	0.000	0.255

Note: One ordinary least-squares regression per process of the per-series $\log_2$ ratio of the MASE of TimesFM-3 to that of AutoARIMA on $\log_2 n$, the version (reference: random parameters with Gaussian innovations) and the drawn parameters, each standardised to mean 0 and standard deviation 1. Heteroskedasticity-robust (HC3) standard errors; 3 200 series per process (3 188 for D8, where series with a zero MASE denominator are excluded). Positive coefficients move the ratio against TimesFM-3.

TABLE S6 | Response-surface regressions of the robustness study, processes D5–D9.

Process	Term	Coefficient	95% interval	<i>p</i>	<i>R</i> ²
D5	version: est period	-0.086	[-0.141, -0.032]	0.002	0.090
D5	version: heavy tail	-0.002	[-0.059, 0.056]	0.957	0.090
D5	version: outliers	-0.151	[-0.209, -0.093]	0.000	0.090
D5	$\log_2 n$	0.147	[0.130, 0.164]	0.000	0.090
D5	parameter sigma	0.032	[0.011, 0.053]	0.003	0.090
D5	parameter amp	-0.021	[-0.042, 0.000]	0.051	0.090
D5	parameter alpha	-0.017	[-0.038, 0.004]	0.107	0.090
D5	parameter beta	-0.008	[-0.029, 0.014]	0.480	0.090
D5	parameter gamma	0.031	[0.010, 0.052]	0.004	0.090
D5	parameter trend	0.010	[-0.010, 0.031]	0.318	0.090
D6	version: est period	-0.002	[-0.058, 0.054]	0.951	0.022
D6	version: heavy tail	0.018	[-0.042, 0.078]	0.558	0.022
D6	version: outliers	-0.023	[-0.081, 0.035]	0.443	0.022
D6	$\log_2 n$	-0.046	[-0.066, -0.027]	0.000	0.022
D6	parameter sigma	-0.029	[-0.052, -0.006]	0.012	0.022
D6	parameter jump sd	-0.038	[-0.058, -0.018]	0.000	0.022
D6	parameter break pos	-0.047	[-0.070, -0.025]	0.000	0.022
D6	parameter sigma level	0.029	[0.009, 0.049]	0.005	0.022
D6	parameter jump sign	-0.011	[-0.032, 0.010]	0.304	0.022
D7	version: est period	-0.002	[-0.081, 0.078]	0.969	0.022
D7	version: heavy tail	-0.031	[-0.116, 0.053]	0.466	0.022
D7	version: outliers	-0.045	[-0.125, 0.036]	0.275	0.022
D7	$\log_2 n$	0.034	[0.010, 0.058]	0.006	0.022
D7	parameter sigma	-0.048	[-0.078, -0.018]	0.002	0.022
D7	parameter r	0.068	[0.038, 0.098]	0.000	0.022
D7	parameter infl pos	-0.081	[-0.113, -0.050]	0.000	0.022
D7	parameter K	0.011	[-0.018, 0.039]	0.474	0.022
D8	version: est period	-0.001	[-0.100, 0.098]	0.985	0.203
D8	version: heavy tail	-0.047	[-0.148, 0.054]	0.361	0.203
D8	version: outliers	-0.025	[-0.125, 0.075]	0.630	0.203
D8	$\log_2 n$	-0.096	[-0.130, -0.062]	0.000	0.203
D8	parameter p	0.509	[0.465, 0.553]	0.000	0.203
D8	parameter lam	0.008	[-0.029, 0.044]	0.685	0.203
D9	version: est period	-0.002	[-0.029, 0.026]	0.905	0.000
D9	version: heavy tail	-0.007	[-0.039, 0.025]	0.677	0.000
D9	version: outliers	-0.013	[-0.041, 0.016]	0.381	0.000
D9	$\log_2 n$	-0.003	[-0.012, 0.007]	0.574	0.000
D9	parameter phi	0.002	[-0.010, 0.014]	0.715	0.000
D9	parameter a	-0.001	[-0.012, 0.010]	0.876	0.000
D9	parameter b	-0.001	[-0.012, 0.010]	0.889	0.000

Note: One ordinary least-squares regression per process of the per-series $\log_2$ ratio of the MASE of TimesFM-3 to that of AutoARIMA on $\log_2 n$, the version (reference: random parameters with Gaussian innovations) and the drawn parameters, each standardised to mean 0 and standard deviation 1. Heteroskedasticity-robust (HC3) standard errors; 3 200 series per process (3 188 for D8, where series with a zero MASE denominator are excluded). Positive coefficients move the ratio against TimesFM-3.